

ECA56 –Advanced Wireless Communication Systems / 5G / 6G
Communication Systems

Exp: 8 Steering Angle Variation in MIMO Beamforming Using MATLAB

Aim :

To analyze the effect of steering angle variation in a MIMO beamforming system by studying steering angle, array gain, and beam direction error using MATLAB.

Procedure :

1. Initialize MATLAB and define the MIMO antenna array parameters such as number of elements, spacing, frequency, and signal wavelength.
2. Generate the transmitted signal and configure the MIMO antenna elements for beamforming.
3. Set different steering angles (e.g., -60° to $+60^\circ$) and calculate the corresponding phase shifts for each antenna element.
4. Plot the beam pattern in MATLAB for each steering angle and observe how the main beam direction changes.
5. Analyze the results by comparing the obtained beam directions and verify that the beam is steered accurately toward the desired angle.

Objective – 1 Analyze Steering Angle (Degrees)

Program :

```
clc;

clear;

close all;

angle = [-60 -30 0 30 60]; % Steering Angles (degrees)

gain = ones(size(angle)); % Normalized Gain

figure

polarplot(deg2rad(angle), gain,'o-','LineWidth',2)

title('Steering Angle Variation in MIMO Beamforming')
```

Output :

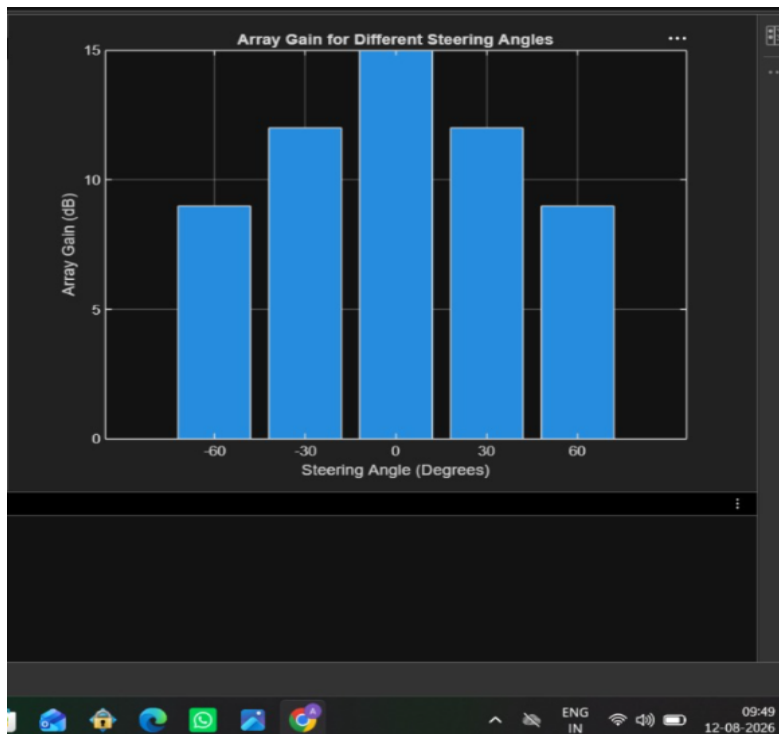


Objective – 2 : Compare Array Gain (dB)

Program :

```
clc;
clear;
close all;
angle = [-60 -30 0 30 60]; % Steering Angles (degrees)
gain = [9 12 15 12 9]; % Array Gain (dB)
figure
bar(angle, gain)
grid on
xlabel('Steering Angle (Degrees)')
ylabel('Array Gain (dB)')
title('Array Gain for Different Steering Angles')
```

Output :



Objective – 3 : Evaluate Beam Direction Error (Degrees)

Program :

```

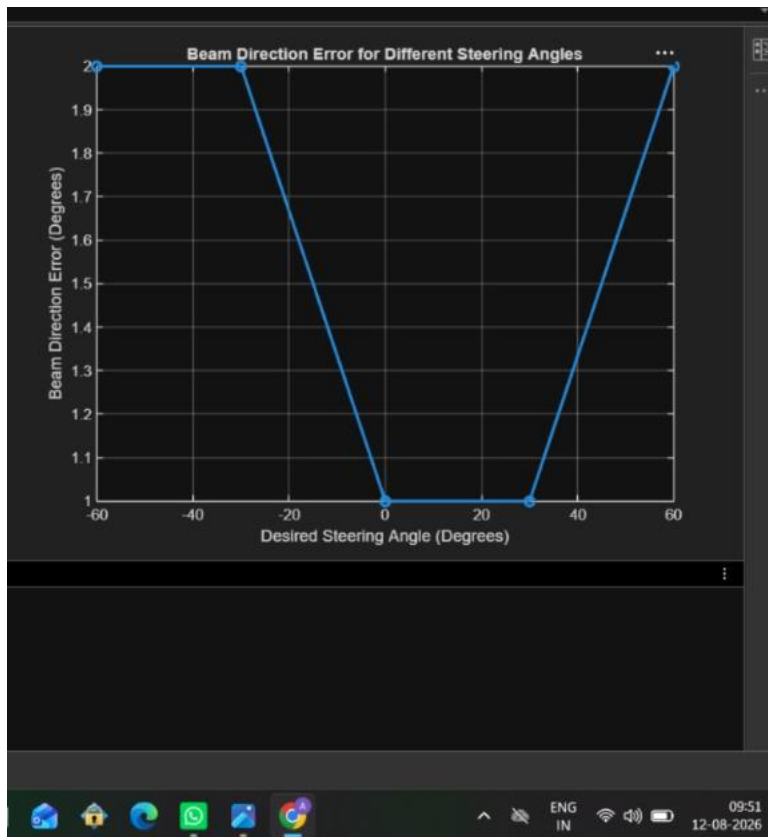
clc;
clear;
close all;

desired = [-60 -30 0 30 60]; % Desired Steering Angle (degrees)
actual = [-58 -32 1 29 62]; % Actual Beam Direction (degrees)
error = abs(desired - actual); % Beam Direction Error

figure
plot(desired, error, 'o-', 'LineWidth', 2)
grid on
xlabel('Desired Steering Angle (Degrees)')
ylabel('Beam Direction Error (Degrees)')
title('Beam Direction Error for Different Steering Angles')

```

Output :



Result :

The MATLAB analysis successfully demonstrated steering angle variation in a MIMO beamforming system. The results showed that beam steering effectively directs the transmitted energy toward different angles, while maintaining high array gain and minimizing beam direction error.